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Coupling between dissimilar rectangular dielectric waveguides

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Abstract

The coupling between two identically excited, dissimilar rectangular dielectric waveguides, whose axes are parallel, is studied as a function of frequency and of the geometrical and the electrical properties of the structure. It is found that, for a given permittivity ratio, the power transfer between the guides is maximal, but not complete, at some particular frequency. It is also found that, for a given frequency, the power transfer between the guides is complete for the appropriate combination of permittivity and size of the guides which achieves the same propagation coefficient in both.